

ANTIMA MODAK

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PROFILE

Software Engineer with 3 years of experience building backend systems, APIs, enterprise integrations, and customer-facing software solutions across SaaS and healthcare. Currently pursuing an MSc in Applied Artificial Intelligence at the University of Warwick, with a focus on practical AI systems, intelligent automation, and enterprise software.

KEY SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL, C++

Frameworks & Libraries: FastAPI, Flask, React, Node.js

Databases: PostgreSQL, MongoDB, MySQL

AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, XGBoost, SHAP, NLP (VADER, LDA, TF-IDF)

Data Science & Visualisation: Pandas, NumPy, Scikit-learn, Streamlit, Plotly

Cloud & DevOps: AWS Lambda, Amazon Cognito, Azure, Docker, Vercel, Render

Tools & Technologies: Git, GitHub, Jira, Confluence, REST APIs, Webhooks, n8n

Software Engineering: REST APIs, Object-Oriented Programming, Data Structures & Algorithms, Agile Development

WORK EXPERIENCE

University of Warwick - Warwick Business School, United Kingdom

March 2026 - Present

Applied AI Analyst (Developer)

- Designed and built a healthcare application end-to-end - architecture, backend, frontend, and deployment
- Partnered directly with the project's physiotherapy team and principal investigator, a recovering Achilles tendon patient and the project's funding lead, to translate lived patient experience into product requirements.
- Developed backend services and REST APIs using Python, Flask, JavaScript, and Supabase, supporting secure data exchange and cloud-based deployment for a physiotherapy patient care use case.
- Leveraged AI-assisted development tools to accelerate prototyping, debugging, and implementation while independently making architectural and technical decisions as the sole engineer on the project.
- Ran early user validation by gathering feedback from real Achilles-injury patients in the physiotherapy team's network ahead of wider rollout.
- Deployed and maintained the application using Git-based CI deployments through Vercel.

TrueLoyal, United Kingdom

October 2025 - Present

Software Engineer (Forward Deployed Engineer)

- Partnered directly with customers and cross-functional stakeholders to scope, design, and deliver technical solutions aligned with business requirements.
- Built REST API and CRM/e-commerce integrations supporting scalable data exchange across client systems.
- Used Docker to containerise and deploy backend services, ensuring consistent environments across client delivery projects.
- Owned projects across the full lifecycle: requirements analysis, implementation, testing, deployment, and support.
- Continued this engagement remotely on a part-time contract basis (via TrueLoyal's San Antonio office) while relocating to the UK for an MSc in Applied AI; title evolved to Forward Deployed Engineer to reflect direct, embedded client-delivery work.

TrueLoyal (Formerly Zinrelo), Pune, India

June 2023 - September 2025

Full Stack Developer

- Collaborated with cross-functional teams and 10+ clients to gather business requirements and deliver customised CRM and loyalty program solutions.
- Developed and deployed over 10 AWS Lambda functions using Node.js, reducing manual workload by over 40% through backend automation.
- Integrated AWS Cognito for secure user login and signup, decreasing authentication-related support tickets by 25%.

- Enhanced the Shopify theme app extension, improving performance and reducing page load time by 35%, leading to higher client satisfaction.
- Designed and implemented 20+ automated workflows with n8n, increasing internal operational efficiency by 30%.
- Participated in code reviews, debugging, and production issue resolution, improving application reliability and software quality.
- Improved data consistency and API response time by 15% through optimisation of backend queries using PostgreSQL and MongoDB.
- Worked with external and internal stakeholders to elicit business requirements and translate them into scalable technical solutions.
- Led requirement-gathering discussions and solution walkthroughs, ensuring alignment between business objectives and technical implementation.

PROJECTS

Enterprise AI Knowledge Platform (LLM/RAG) | github

July 2026

- Designed and built a full-stack Retrieval-Augmented Generation (RAG) platform enabling secure querying of enterprise documents with source citations, audit logging, and role-based access control.
- Implemented document ingestion, chunking, embedding generation, vector similarity search, and LLM-powered answer generation using Ollama for local development and Gemini for cloud deployment.
- Developed JWT authentication, query history, and audit logging to provide enterprise-style governance and document traceability.
- Developed a React frontend and FastAPI backend with PostgreSQL and pgvector for semantic document retrieval.
- Containerised and deployed the application using Docker, Render, and Vercel.

Explainable AI Framework for Supply Chain Return Analysis

June 2026 - July 2026

- Built a 10-phase AI pipeline classifying product return root causes with 97.1% accuracy (XGBoost, Macro F1: 0.964, AUC: 0.999).
- Applied an NLP pipeline (VADER, Gensim LDA, TF-IDF) to 50,000 customer reviews to extract structured features driving model predictions.
- Implemented SHAP explainability, making every ML prediction auditable for supply chain managers.
- Engineered a composite risk scoring framework quantifying £308,000 in annual return cost exposure.
- Integrated a locally-run LLM (Ollama, Llama 3.2) to generate stakeholder-specific natural language recommendations, delivered via a Streamlit dashboard with scenario simulation.

VTap: AI-powered Payment Optimisation Tool

March 2026

- Selected as a Top 4 finalist (out of 26 teams) and presented at Google London.
- Developed an AI-assisted payment optimisation platform using transaction analytics.
- Designed data-driven decision workflows to improve payment success rates.
- Presented the solution to judges from Google and NatWest.

EDUCATION

The University of Warwick, WMG | United Kingdom

September 2025 - Present

MSc Applied Artificial Intelligence (Expected Aug 2026)

Relevant Modules: Artificial Intelligence & Deep Learning, Programming for AI, Applied Statistics for AI, Data Mining, AI for Industry, Ethical AI, Dissertation: Explainable AI Framework for Supply Chain Return Analysis

Vishwakarma Institute of Technology | India

August 2019 - June 2023

B.Tech in Computer Engineering

Leadership & Engagement

Student Ambassador

September 2025 - Present

University of Warwick, WMG

- Represented WMG at university open days and postgraduate recruitment events, engaging prospective students and sharing firsthand experience of the MSc Applied Artificial Intelligence programme.
- Delivered live presentations and Q&A sessions to prospective students and families, building public speaking and cross-audience communication skills.